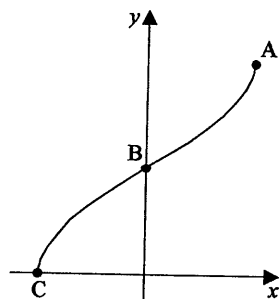


QUESTION 1

- (a) Solve the inequality $\frac{1}{x} < \frac{1}{x+1}$. 3
- (b) (i) When a polynomial is divided by a quadratic, what is the most general form of the remainder? 1
- (ii) The remainder when $P(x)$ is divided by $(x-2)$ is 4. The remainder when $P(x)$ is divided by $(x-3)$ is 9. Find the remainder when $P(x)$ is divided by $(x-2)(x-3)$. 2
- (c) Use the substitution $u = 2 - x$, to evaluate $\int_{-1}^2 x\sqrt{2-x} dx$. 4
- (d) Find $\int \sin^2 x dx$. 2

QUESTION 2

- (i) Find $\lim_{x \rightarrow 0} \frac{\sin \frac{x}{3}}{x}$. 1
- (ii) If $f(x) = e^{x+2}$,
 (i) Find the inverse function $f^{-1}(x)$. 2
 (ii) State the domain and range of $f^{-1}(x)$. 2
 (iii) On one diagram sketch the graphs of $f(x)$ and $f^{-1}(x)$. 2
- (iii) The diagram below shows the graph of $y = \pi + 2 \sin^{-1} 3x$.



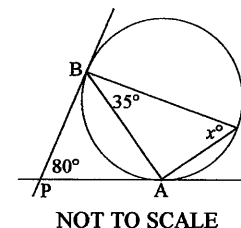
- (i) Write down the co-ordinates of the endpoints A and C. 2
- (ii) Write down the co-ordinates of the point B. 1
- (iii) Find the equation of the tangent to the curve $y = \pi + 2 \sin^{-1} 3x$ at the point B. 2

QUESTION 3

- (a) The equation $x^3 - 2x^2 + 4x - 5 = 0$ has roots α, β, γ .
 Find the values of:
 (i) $\alpha\beta\gamma$ 1
 (ii) $\alpha\beta + \beta\gamma + \gamma\alpha$ 1
 (iii) $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$. 2
- (b) By using the expansion of $\tan(\alpha + \beta)$ find the value of k such that
 $\tan^{-1}(k) + \tan^{-1}\left(\frac{2}{3}\right) = \frac{\pi}{4}$ 3
- (c) (i) Express $\sqrt{3} \cos \theta - \sin \theta$ in the form $A \cos(\theta + \alpha)$ 2
 (ii) Solve the equation $\sqrt{3} \cos \theta - \sin \theta = 1$ for $0 \leq \theta \leq 2\pi$ 2
 (iii) What is the general solution of the equation? 1

QUESTION 4

- (a) 3

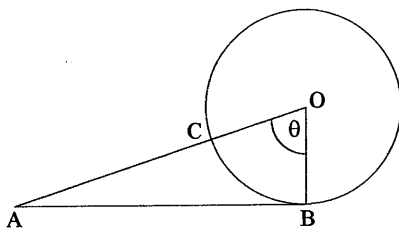


PA and PB are tangents to the circle. Find the value of x giving reasons for your answer.

- (b) A man standing 80 metres from the base of a high-rise building observes an external lift moving up outside the building at a constant rate of 7 metres per second.
 (i) If θ radians is the angle of elevation of the lift from the observer, find an expression for $\frac{d\theta}{dt}$ in terms of θ . 3
 (ii) Evaluate $\frac{d\theta}{dt}$ at the instant when the lift is 30 metres above the observer's horizontal line of vision. Give your answer to 2 significant figures. 1
- (c) The speed v centimetres/second of a particle moving with simple harmonic motion in a straight line is given by $v^2 = 6 + 4x - 2x^2$, where x cm is the magnitude of the displacement from a fixed point O.
 (i) Show that $\ddot{x} = -2(x - 1)$. 2
 (ii) Find the centre of the motion. 1
 (iii) Find the period of the motion. 1
 (iv) Find the amplitude of the motion. 1

QUESTION 5

- (i) Differentiate $x \cos^{-1}x - \sqrt{1-x^2}$ 2
- (ii) Hence evaluate $\int_0^1 \cos^{-1}x \, dx$ 2
- Find:
- (i) $\int \frac{x}{3+4x^2} \, dx$ 1
- (ii) $\int \frac{dx}{3+4x^2}$ 2



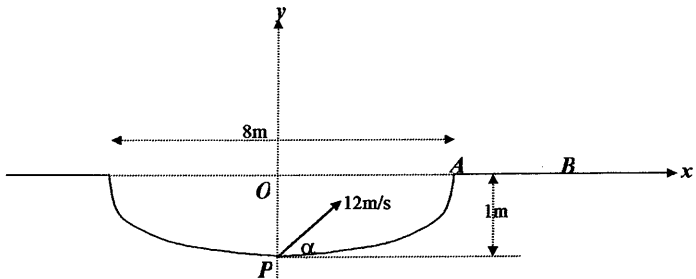
- In the above diagram, O is the centre of a circle and AB is a tangent to the circle, meeting it at point B. The line interval OA cuts the circumference of the circle at a point C.
- (i) If the arc of the circle CB divides the triangle AOB into two portions of equal area and if the angle AOB is denoted by θ , show that $\tan \theta = 2\theta$. 2
- (ii) If $\theta = 1.2$ radians is an approximate solution to the equation in (i) above, use one application of Newton's Method to find a better approximation, correct to two decimal places. 3

QUESTION 6

- Use Mathematical Induction to show that $\cos(x + n\pi) = (-1)^n \cos x$ for all positive integers $n \geq 1$. 4
- P($2ap, ap^2$) is a point on the parabola $x^2 = 4ay$.
- (i) Show that the equation of the normal to the curve of the parabola at the point, P, is: $x + py = 2ap + ap^3$. 2
- (ii) Find the co-ordinates of the point Q where the normal at P meets the y axis. 2
- (iii) Determine the co-ordinates of the point R which divides PQ externally in the ratio 2 : 1. 2
- (iv) Find the cartesian equation of the locus of R and describe the locus in geometrical terms. 2

QUESTION 7

- (a) Prove that $\frac{1 + \sin \theta - \cos \theta}{1 + \sin \theta + \cos \theta} = \tan\left(\frac{\theta}{2}\right)$ 4
- (b) A golf ball is lying at point P, at the middle of a sand bunker, which is surrounded by level ground. The point A is at the edge of the bunker, and line AB lies on level ground. The bunker is 8 metres wide and 1 metre deep.
- The ball is hit towards A with an initial speed of 12 metres per second, and angle of elevation α . You may assume that the acceleration due to gravity is 10m/s^2 .



- (i) Show that the golf ball's trajectory at time t seconds after being hit is defined by the equations : 2
- $x = 12t \cos \alpha$ and $y = -5t^2 + 12t \sin \alpha - 1$
- where x and y are the horizontal and vertical displacements in metres of the ball from the origin O shown in the diagram.
- (ii) Given $\alpha = 30^\circ$, how far from A will the ball land? 2
- (iii) Find the maximum height above the ground reached by the ball if $\alpha = 30^\circ$. 1
- (iv) Find the range of values of α , to the nearest degree, at which the ball must be hit so that it will land to the right of A. 3

QUESTION 1

a) $\frac{1}{x} < \frac{1}{x+1}$

Boundary pts.
 $x=0, x=-1$

Let $\frac{1}{x} = \frac{1}{x+1}$
 $x+1 = x$
No solution



$$-1 < x < 0$$

OR

$$x(x+1)^2 < x^2(x+1)$$

$$x^3 + 2x^2 + x < x^3 + x^2$$

$$x^2 + x < 0$$

$$x(x+1) < 0$$

$$-1 < x < 0$$

b) i) $ax+b$, a and b constants

ii) $P(x) = (x-2)(x-3)Q(x) + ax+b$

$P(2) = 2a+b = 4$

$P(3) = 3a+b = 9$

$a=5, b=-6$

Remainder $5x-6$

c) $\int_{-1}^2 x\sqrt{2-x} dx$

$$u = 2-x$$

$$du = -dx$$

$$\text{When } x=-1, u=3$$

$$x=2, u=0$$

$$= -\int_3^0 (2-u)\sqrt{u} du$$

$$= \int_0^3 2u^{1/2} - u^{3/2} du$$

$$= \left[2 \cdot \frac{2}{3} u^{3/2} - \frac{2}{5} u^{5/2} \right]_0^3$$

$$= 4\sqrt{3} - \frac{2}{5} 9\sqrt{3}$$

$$= \frac{2\sqrt{3}}{5}$$

$$d) \int \sin^2 x dx = \frac{1}{2} \int (1 - \cos 2x) dx$$

$$= \frac{1}{2} \left[x - \frac{1}{2} \sin 2x \right] + C$$

$$= \frac{x}{2} - \frac{1}{4} \sin 2x + C$$

QUESTION 2

$$a) \lim_{x \rightarrow 0} \frac{\sin \frac{x}{3}}{x} = \lim_{x \rightarrow 0} \frac{\sin \frac{x}{3}}{\frac{x}{3} \cdot 3} = \frac{1}{3} \quad (1)$$

b) (i) $f(x) = e^{x+2}$

for inverse: $x = e^{y+2}$

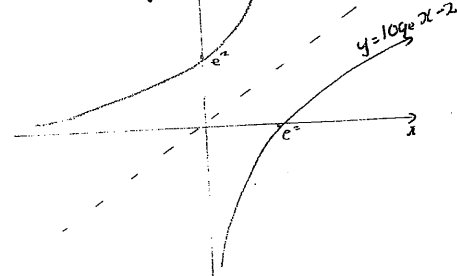
$\log_e x = y+2$

$y = \log_e x - 2$

$f^{-1}(x) = \log_e x - 2$

D: $x > 0$ (1)

R: all real y (1)



(2)

c) i) C: $y=0, 0 = \pi + 2\sin^{-1} 3x$

$\sin^{-1} 3x = -\frac{\pi}{2}$

$3x = -1$

$x = -\frac{1}{3}$

$C(-\frac{1}{3}, 0)$

A: $x = \frac{1}{3}, y = \pi + 2\sin^{-1} 1$

$= 2\pi$

$A(\frac{1}{3}, 2\pi)$

ii) B (0, π)

iii) $y = \pi + 2\sin^{-1} 3x$

$\frac{dy}{dx} = 2 \cdot \frac{1}{\sqrt{1-9x^2}} \cdot 3$

$\frac{dy}{dx} = \frac{6}{\sqrt{1-9x^2}}$

Eqn: $y - \pi = 6x$

$6x - y + \pi = 0$

(2)

QUESTION 3

a) $x^3 - 2x^2 + 4x - 5 = 0$

i) $\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a} = 4$

$\alpha\beta\gamma = -\frac{d}{a} = 5$

$$\text{ii) } \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \beta\gamma + \gamma\alpha}{\alpha\beta\gamma}$$

$$= \frac{4}{5}$$

b) $\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$

$$\tan(\tan^{-1} k + \tan^{-1} \frac{2}{3}) = \frac{k + \frac{2}{3}}{1 - \frac{2}{3}k}$$

$\tan \frac{\pi}{4} = 1$

$\frac{k + \frac{2}{3}}{1 - \frac{2}{3}k} = 1$

$k + \frac{2}{3} = 1 - \frac{2}{3}k$

$\frac{5}{3}k = \frac{1}{3}$

$k = \frac{1}{5}$

$$c) \text{ i) } \sqrt{3} \cos \theta - \sin \theta = A \cos(\theta + \alpha)$$

$$= A \cos \theta \cos \alpha - A \sin \theta \sin \alpha$$

$A \cos \alpha = \sqrt{3}$

$A \sin \alpha = 1$

$\tan \alpha = \frac{1}{\sqrt{3}}, \alpha = \frac{\pi}{6}$

$A = 2$

$\sqrt{3} \sin \theta - \sin \theta = 2 \cos(\theta + \frac{\pi}{6})$

ii) $2 \cos(\theta + \frac{\pi}{6}) = 1$

$\cos(\theta + \frac{\pi}{6}) = \frac{1}{2}$

$\theta + \frac{\pi}{6} = \frac{\pi}{3}, \frac{5\pi}{3}$

$\theta = \frac{\pi}{6}, \frac{3\pi}{2}$

iii) $\theta = \frac{\pi}{6} \pm 2n\pi, \frac{3\pi}{2} \pm 2n\pi$

QUESTION 4

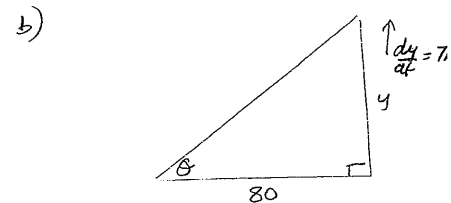
4) a) $PA = PB$ (tangents from external pt. are =)

$\therefore \angle PBA = \angle PAB$ (base $\angle$ s of isos.)

$$\therefore \angle PAB = \frac{180 - 80}{2} \text{ (sum of } \angle \text{s of } \triangle \text{)}$$

$$= 50^\circ$$

$\therefore \angle P = 50^\circ$ (Alternate segment)



$\tan \theta = \frac{y}{80}$

$y = 80 \tan \theta$

$\frac{dy}{d\theta} = 80 \sec^2 \theta$

$\frac{d\theta}{dt} = \frac{d\theta}{dy} \cdot \frac{dy}{dt}$

$= \frac{1}{80 \sec^2 \theta} \cdot 7$

$= \frac{7 \cos^2 \theta}{80}$

When $y = 30, \tan \theta = \frac{3}{8}$

$\theta \approx 0.3587$

$\frac{d\theta}{dt} \approx 0.077 \text{ rad/sec}$

c) $v^2 = 6 + 4x - 2x^2$

i) $\dot{x} = \frac{d}{dx} (\frac{1}{2} v^2)$

$= \frac{d}{dx} (3 + 2x - x^2)$

$= 2 - 2x$

$= -2(x-1)$

ii) Centre of motion $\dot{x} = 0$

$x = 1$

iii) Period $= \frac{2\pi}{n} = \frac{2\pi}{\frac{1}{2}}$

iv) $v = 0$

$x^2 - 2x - 3 = 0$

$(x-3)(x+1) = 0$

Extremities $x = 3, -1$, Centre $x = 1$

Amplitude is 2cm

QUESTION 5

a) i) $\frac{d}{dx} (x \cos^{-1} x - \sqrt{1-x^2})$
 $= 1 \cdot \cos^{-1} x - x \cdot \frac{1}{\sqrt{1-x^2}} - \frac{1}{2} \frac{1-x^2}{\sqrt{1-x^2}}$

i for $\frac{d}{dx} (x \cos^{-1} x)$
 i for $\frac{d}{dx} (\sqrt{1-x^2})$ (2)

ii) $\int_0^1 \cos^{-1} x \, dx = \left[x \cos^{-1} x - \sqrt{1-x^2} \right]_0^1$
 $= \cos^{-1} 1 - (-1)$
 $= 0 + 1$
 $= 1$

b) i) $\int \frac{x}{3+4x^2} \, dx = \frac{1}{8} \ln(3+4x^2) + C$

ii) $\int \frac{dx}{3+4x^2} = \int \frac{dx}{(\sqrt{3})^2 + (2x)^2}$
 $= \frac{1}{2} \cdot \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2x}{\sqrt{3}} \right) + C$
 $= \frac{1}{2\sqrt{3}} \tan^{-1} \left(\frac{2x}{\sqrt{3}} \right) + C$

c) i) Area $\triangle AOB = 2 \times$ Area sector OAB

Let $OB = r$, $\frac{AB}{r} = \tan \theta$

$\frac{1}{2} OB \cdot AB = 2 \times \frac{1}{2} r^2 \theta$

$\frac{1}{2} r \cdot r \tan \theta = r^2 \theta$
 $\therefore \tan \theta = 2\theta$

ii) $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$

$f(x) = \tan \theta - 2\theta$

$f'(x) = \sec^2 \theta - 2$

$f(1.2) \doteq 0.17215$

$f'(1.2) \doteq 5.61596$

$x_2 = 1.2 - \frac{0.17215}{5.61596}$

$\doteq 1.17$ correct to 2 dec. places

QUESTION 6

a) Prove $\cos(x + n\pi) = (-1)^n \cos x$

① If $n=1$
 L.H.S. $= \cos(\pi + x)$
 $= -\cos x$

R.H.S. $= (-1)^1 \cos x$
 $= -\cos x$

$\therefore$ True for $n=1$.

② Assume true for $n=k$.

$\cos(x + k\pi) = (-1)^k \cos x$

③ If $n=k+1$, R.T.P.

$\cos(x + k\pi + \pi) = (-1)^{k+1} \cos x$

L.H.S. $= \cos(\pi + x + k\pi)$

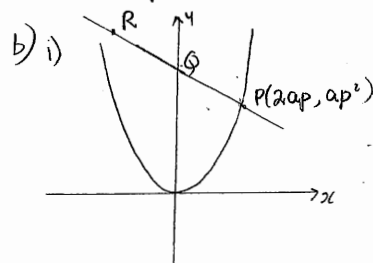
$= -\cos(x + k\pi)$

$= -(-1)^k \cos x$ from ②

$= (-1)^{k+1} \cos x$

④ Since it is true for $n=1$, it is true for $n=1+1$, and then true for $n=2+1$ and so on.

$\therefore$ True for all n



$y = \frac{x^2}{4a}$

$\frac{dy}{dx} = \frac{x}{2a}$ at $P(2ap, ap^2)$, $m_T = P$

$m_N = -\frac{1}{P}$

Eqn of normal: $y - ap^2 = -\frac{1}{P}(x - 2ap)$

$-py + ap^3 = x - 2ap$
 $x + py = 2ap + ap^3$

ii) Q. $x=0$
 $\therefore py = 2ap + ap^3$
 $y = 2a + ap^2$
 $Q(0, 2a + ap^2)$

iii) $P(2ap, ap^2)$ $Q(0, 2a + ap^2)$
 $2 \therefore -1$

$R\left(\frac{2x_0 + -1x_2ap}{2}, \frac{2(2a + ap^2) + -ap^2}{2}\right)$

$R(-2ap, 4a + ap^2)$

n) $x = -2ap$

$P = \frac{x}{-2a}$

$y = 4a + a\left(\frac{x}{-2a}\right)^2$
 $= 4a + \frac{x^2}{4a}$

$\therefore x^2 = 4a(y - 4a)$

Parabola, focal length a , Vertex $(0, 4a)$

Question 7

a) Let $t = \tan \frac{\theta}{2}$

$$\text{L.H.S} = \frac{1 + \sin \theta - \cos \theta}{1 + \sin \theta + \cos \theta}$$

$$= 1 + \frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2}$$

$$\frac{1 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}}$$

$$= \frac{1+t^2+2t-1+t^2}{1+t^2+2t+1-t^2}$$

$$= \frac{2t^2+2t}{2+2t}$$

$$= \frac{2t(t+1)}{2(1+t)}$$

$$= t$$

2

(4)

b) i) $\dot{x} = 0$

$$\dot{x} = C_1$$

When $t=0$, $\dot{x} = 12 \cos \alpha$

$$\dot{x} = 12 \cos \alpha$$

$$x = 12t \cos \alpha + C_2$$

When $t=0$, $x=0$

$$\therefore x = 12t \cos \alpha \quad (1) \text{ from above}$$

ii) $\alpha = 30^\circ$

Ball lands when $y=0$

$$\therefore 12t \frac{1}{2} - 5t^2 - 1 = 0$$

$$\therefore 6t - 5t^2 - 1 = 0$$

$$5t^2 - 6t + 1 = 0$$

$$(5t-1)(t-1) = 0$$

$$t = \frac{1}{5}, \quad t = 1$$

ball lands on top

$$\ddot{y} = -10$$

$$\dot{y} = -10t + C_3$$

When $t=0$, $\dot{y} = 12 \sin \alpha$

$$\therefore \dot{y} = 12 \sin \alpha - 10t$$

$$y = 12t \sin \alpha - 5t^2 + C_4$$

When $t=0$, $y=-1$

$$\therefore y = 12t \sin \alpha - 5t^2 - 1$$

(1) from above

(2)

i) When $t=1$

$$x = 12 \cos 30^\circ = 6\sqrt{3}$$

$$\therefore \text{Distance from A} = 6\sqrt{3} - 4 = 6.4 \text{ m}$$

(2)

ii) Max. Height. $\dot{y} = 0$

$$\therefore t = \frac{6 \sin \alpha}{5}, \quad \alpha = 30^\circ = \frac{3}{5}$$

(1)

iv) $x = 12t \cos \alpha$

$$t = \frac{x}{12 \cos \alpha}$$

$$\therefore y = -5 \left(\frac{x}{12 \cos \alpha} \right)^2 + 12 \left(\frac{x}{12 \cos \alpha} \right) \sin \alpha - 1$$

$$= -\frac{5}{144} x^2 \sec^2 \alpha + x \tan \alpha - 1$$

$$x=4, \quad y=0$$

$$\therefore -\frac{5}{9} \sec^2 \alpha + 4 \tan \alpha - 1 = 0$$

$$5(\tan^2 \alpha + 1) - 36 \tan \alpha + 9 = 0$$

$$5 \tan^2 \alpha - 36 \tan \alpha + 14 = 0$$

$$\tan \alpha = \frac{36 \pm \sqrt{36^2 - 4 \times 5 \times 14}}{10}$$

$$= 0.4125, \quad 6.7875$$

$$\alpha = 22.4^\circ, \quad 81.6^\circ$$

$\therefore$ The required range is

$$23^\circ < \alpha < 81^\circ$$

(3)